Chapter 7 Assignment — Analytic Trigonometry

Name: ____________________ Date: ____________________

How to use this set: Each topic begins with *Strategy Notes* and a *Worked Example*. Then try the *Practice* (skill) and the *Challenge* (deeper) items. Give exact answers using radicals and π when possible. State domain restrictions when solving equations/inequalities.

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1. **Topic 1: Core Trigonometric Identities**

   Strategy Notes. Use the Pythagorean identities $\sin^2 x + \cos^2 x = 1$, $1 + \tan^2 x = \sec^2 x$, and $1 + \cot^2 x = \csc^2 x$. Convert everything to sines and cosines when stuck, factor common terms, and avoid cross-multiplying until both sides are simplified.

   Worked Example. Verify the identity $\tan\theta \sin\theta + \cos\theta = \sec\theta$.

   Solution. $\tan\theta \sin\theta = \dfrac{\sin\theta}{\cos\theta}\sin\theta = \dfrac{\sin^2\theta}{\cos\theta}$. Then $\dfrac{\sin^2\theta}{\cos\theta} + \cos\theta = \dfrac{\sin^2\theta + \cos^2\theta}{\cos\theta} = \dfrac{1}{\cos\theta} = \sec\theta$.

   Practice.

   (a) Prove the identity $\sin x \tan\left(\frac{x}{2}\right) = 1 - \cos x$.

   (b) Show that $\cos 4x = 1 - 8\sin^2 x + 8\sin^4 x$.

Challenge.

(a) Verify $\dfrac{\cot x - 1}{\cot x + 1} = \dfrac{\cos 2x}{1 + \sin 2x}$.

2. **Topic 2: Addition and Subtraction Formulas**

   Strategy Notes. $\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$ and $\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta$. Use these to rewrite linear combinations of $\sin x$ and $\cos x$, and to evaluate nonstandard angles.

   Worked Example. Simplify $\sin(x-y)\cos y + \cos(x-y)\sin y$.

   Solution. By the sine addition formula, this equals $\sin\big((x-y)+y\big) = \sin x$.

   Practice.

   (a) Rewrite $\sqrt{2}\cos\big(\frac{\pi}{4} - x\big)$ in the form $A\cos x + B\sin x$ and give A, B.

   (b) Express $\cos(\theta - \pi)$ entirely in terms of $\sin\theta$ and/or $\cos\theta$.

Challenge.

(a) Evaluate $\tan 285^\circ$ exactly.

3. **Topic 3: Double-, Half-Angle, and Product-to-Sum Formulas**

   Strategy Notes. Use $\sin 2x = 2\sin x\cos x$, $\cos 2x = \cos^2 x - \sin^2 x = 1 - 2\sin^2 x = 2\cos^2 x - 1$. Half-angle: $\sin^2\left(\frac{x}{2}\right) = \frac{1-\cos x}{2}$ and $\cos^2\left(\frac{x}{2}\right) = \frac{1+\cos x}{2}$. Product-to-sum converts products like $\sin ax \sin bx$ to sums.

   Worked Example. Compute $\sin\left(\frac{\pi}{12}\right)$ exactly.

   Solution. $\frac{\pi}{12} = \frac{\pi}{3} - \frac{\pi}{4}$. Then $\sin(\frac{\pi}{3} - \frac{\pi}{4}) = \sin\frac{\pi}{3}\cos\frac{\pi}{4} - \cos\frac{\pi}{3}\sin\frac{\pi}{4} = \frac{\sqrt{3}}{2}\cdot\frac{\sqrt{2}}{2} - \frac{1}{2}\cdot\frac{\sqrt{2}}{2} = \frac{\sqrt{6}-\sqrt{2}}{4}$.

   Practice.

   (a) If $\sin 2x = \dfrac{24}{25}$ with $0 < x < \frac{\pi}{2}$, find $\sin^4 x + \cos^4 x$.

   (b) Find the exact value of $\sin 75°$.

Challenge.

(a) Determine constants a, b so that $\dfrac{-8+9\cos^2\theta}{1-3\sin\theta} = a + b\sin\theta$.

4. **Topic 4: Solving Trigonometric Equations and Inequalities**

   Strategy Notes. Isolate a basic trig function, use identities to factor/rewrite, and check that solutions lie in the stated interval. For inequalities, analyze sign charts or transform to equivalent algebraic inequalities on reference angles.

   Worked Example. Solve $\sin 2x = \sin x$ for $0 \leq x \leq 2\pi$.

   Solution. $\sin 2x - \sin x = \sin x(2\cos x - 1) = 0$. Thus $\sin x = 0$ gives $x = 0, \pi, 2\pi$, and $2\cos x - 1 = 0$ gives $\cos x = \frac{1}{2}$, i.e., $x = \frac{\pi}{3}, \frac{5\pi}{3}$.

   Practice.

   (a) Find all x with $0 \leq x \leq 2\pi$ such that $\sin 2x \leq \sin x$.

   (b) For $-\pi \leq \theta \leq \pi$, solve $2(\sin^2\theta - \cos^2\theta) = 8\sin\theta - 5$.

Challenge.

(a) Solve $\cos 3x = \sin x$ for $0 \leq x < 2\pi$.

5. **Topic 5: Phase-Amplitude Form and Mixed Skills**

   Strategy Notes. Any expression of the form $A\sin t + B\cos t$ can be written as $R\sin(t+\varphi)$ where $R = \sqrt{A^2+B^2}$ and $\tan\varphi = \dfrac{B}{A}$. This makes ranges and extrema immediate.

   Worked Example. Find the range of $g(t) = 2\sin t + 3\cos t$.

   Solution. Write $2\sin t + 3\cos t = R\sin(t+\varphi)$ with $R = \sqrt{2^2+3^2} = \sqrt{13}$. Hence $-\sqrt{13} \le g(t) \le \sqrt{13}$.

   Practice.

   (a) If $\cos\theta = -0.6$, find all possible values of $\sin 2\theta$.

   (b) If $\sin x + \cos x = \dfrac{1}{5}$ and $\frac{\pi}{2} < x < \pi$, find $\tan x$.

Challenge.

(a) Suppose $\sin a + \sin b = \sqrt{\frac{5}{3}}$ and $\cos a + \cos b = 1$. Find $\cos(a - b)$.